Control of Mobile Robots - Laboratory session 1
Introduction to C++
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Develop two C++ classes: one implementing a PID controller and the other implementing a first-order dynamic system.
The **PID controller** class shall provide methods to:

- construct and initialize the controller;
- update the reference and measured signals;
- compute the control signal;
- retrieve the current value of the control signal.

Similarly, the **first-order system** class shall provide methods to:

- construct and initialize the system;
- update the input signal;
- compute the output signal;
- retrieve the current value of the output signal.

Provide two separate main functions to independently test the behaviour of the PID controller and the first-order system.
Finally, provide a third main function that:

- creates a PI controller with proportional gain $K_C = 1.0$, integral time $T_I = 10.0$ s, and sampling time $T_s = 0.1$ s;
- creates a first-order system with gain $K_P = 10.0$, time constant $T = 10.0$ s, and sampling time $T_s = 0.1$ s;
- connects the controller and the first-order system in a closed-loop configuration;
- applies a step reference and evaluates the resulting closed-loop response.

Use the resulting step response to verify the behaviour of the closed-loop system.

The PID controller can be discretized using the backward (implicit) Euler method, resulting in the following update equations:

$$\begin{aligned} u_{I_k} &= u_{I_{k-1}} + K_C \frac{T_s}{T_I} e_k \\ u_{D_k} &= \frac{T_D}{N T_s + T_D} u_{D_{k-1}} + K_C N \left(e_k - e_{k-1}\right) \\ u_k &= K_C e_k + u_{I_k} + u_{D_k} \end{aligned}$$

Here, K_C, T_I, T_D, and N are the PID controller parameters, and T_s is the sampling time. The quantities e_k, u_{I_k}, u_{D_k}, and u_k denote, respectively, the control error, the integral contribution, the derivative contribution, and the controller output at time step k.
Similarly, consider a first-order system described by the transfer function

$$G(s) = \frac{K_P}{1 + sT}$$

Using the backward (implicit) Euler method, its discrete-time update equation is

$$y_k = \frac{T}{T + T_s} y_{k-1} + K_P \frac{T_s}{T + T_s} u_k.$$

Here, K_P is the system gain, T is the time constant, and u_k and y_k denote, respectively, the input and output signals at time step k.